

Calibration results

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Normalized Residuals

Reprojection error (cam0): mean 4.05624277749, median 3.37080443656, std: 2.7888140625
Gyroscope error (imu0): mean 2.99678743208, median 1.93309953614, std: 3.73673416869
Accelerometer error (imu0): mean 5.19139426495, median 4.32422259221, std: 4.75256735221

Residuals

Reprojection error (cam0) [px]: mean 4.05624277749, median 3.37080443656, std: 2.7888140625
Gyroscope error (imu0) [rad/s]: mean 0.00131435856564, median 0.000847836555357, std: 0.00163889120381
Accelerometer error (imu0) [m/s²]: mean 0.388291084408, median 0.323430853805, std: 0.355468962042

Transformation (cam0):

T_ci: (imu0 to cam0):

```
[[ -0.01111141 -0.99560383 -0.09300291 -0.03682145]
 [ 0.00281782 0.09297711 -0.99566426 0.03659548]
 [ 0.9999343 -0.0113253 0.00177233 -0.22148021]
 [ 0.         0.         0.         1.        ]]
```

T_ic: (cam0 to imu0):

```
[[ -0.01111141 0.00281782 0.9999343 0.2209534 ]
 [ -0.99560383 0.09297711 -0.0113253 -0.04257045]
 [ -0.09300291 -0.99566426 0.00177233 0.03340484]
 [ 0.         0.         0.         1.        ]]
```

timeshift cam0 to imu0: [s] ($t_{imu} = t_{cam} + \text{shift}$)

-0.0221861416145

Gravity vector in target coords: [m/s²]

[1.4121539 -9.68847438 -0.55471478]

Calibration configuration

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cam0

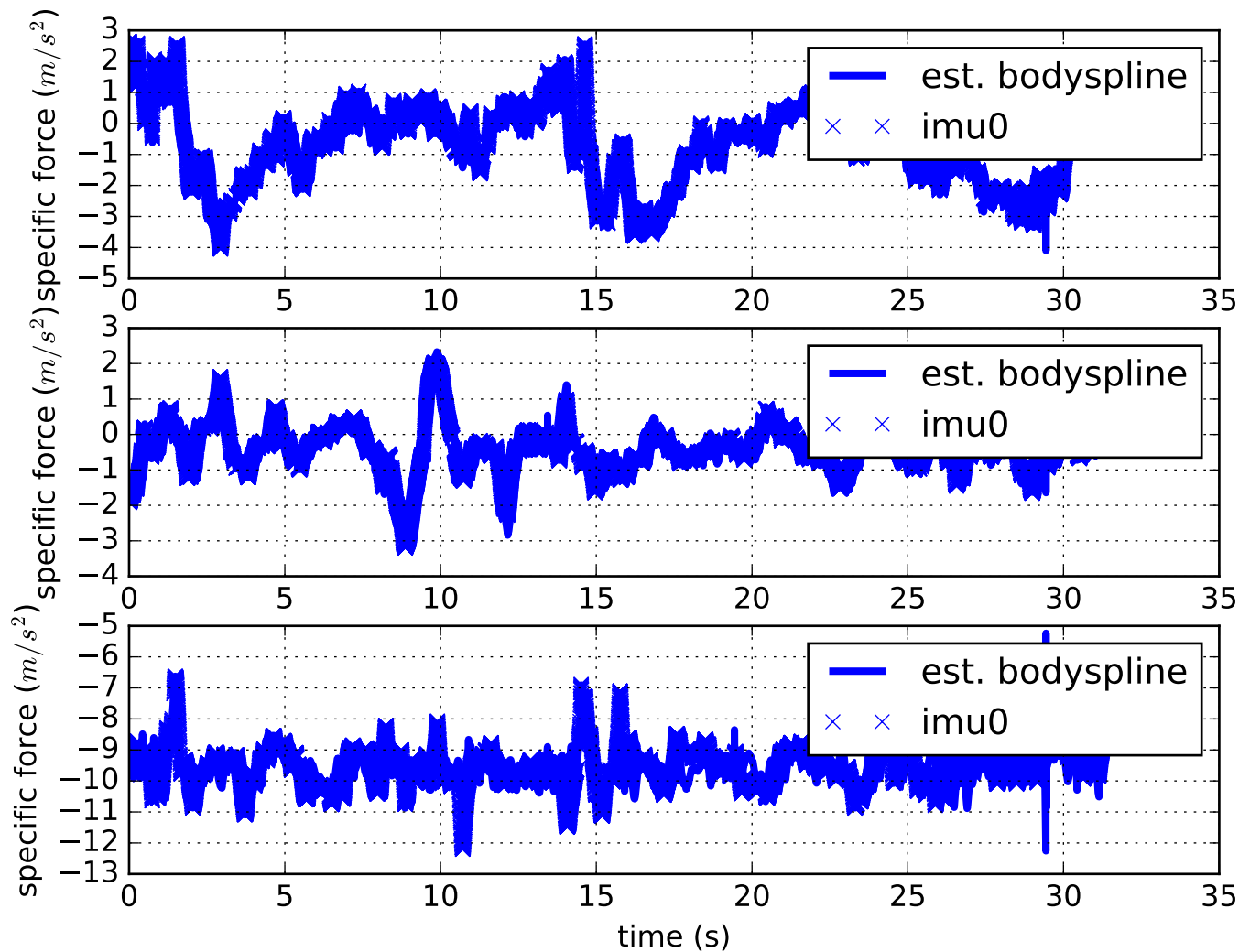
Camera model: pinhole
Focal length: [567.2599929113292, 567.3520242996028]
Principal point: [653.1173010554967, 398.8548950494993]
Distortion model: equidistant
Distortion coefficients: [-0.04435617597702202, 0.01110805569803823, -0.016260141169523998, 0.005140841898676915]
Type: aprilgrid
Tags:
 Rows: 6
 Cols: 6
 Size: 0.088 [m]
 Spacing 0.0264 [m]

IMU configuration
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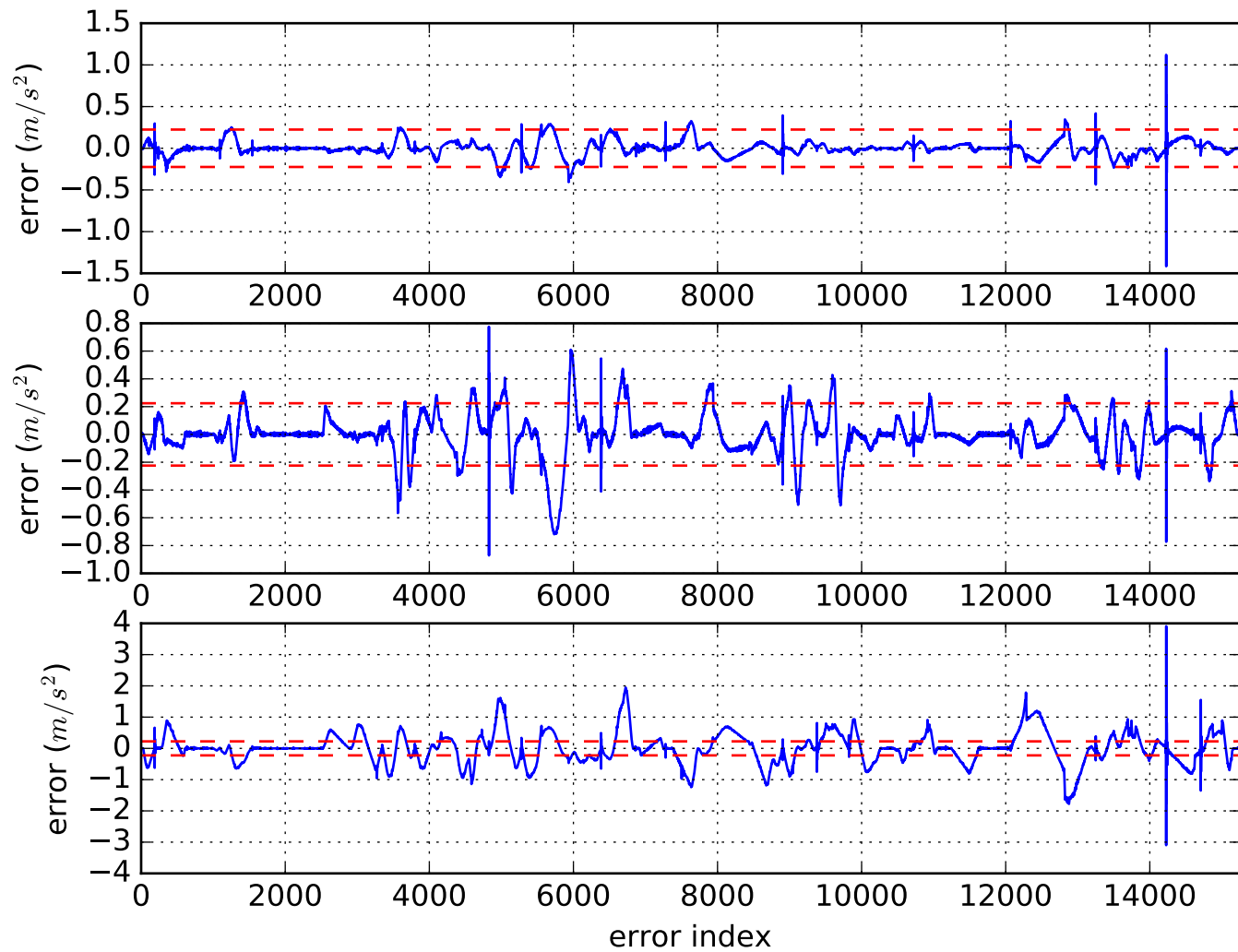
IMU0:

Model: calibrated
Update rate: 481
Accelerometer:
 Noise density: 0.00341036503941
 Noise density (discrete): 0.0747951445394
 Random walk: 0.00424360641439
Gyroscope:
 Noise density: 1.99979456114e-05
 Noise density (discrete): 0.00043858918773
 Random walk: 9.23535168218e-08
 T_{j_b}
 [[1. 0. 0. 0.]
 [0. 1. 0. 0.]
 [0. 0. 1. 0.]
 [0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]

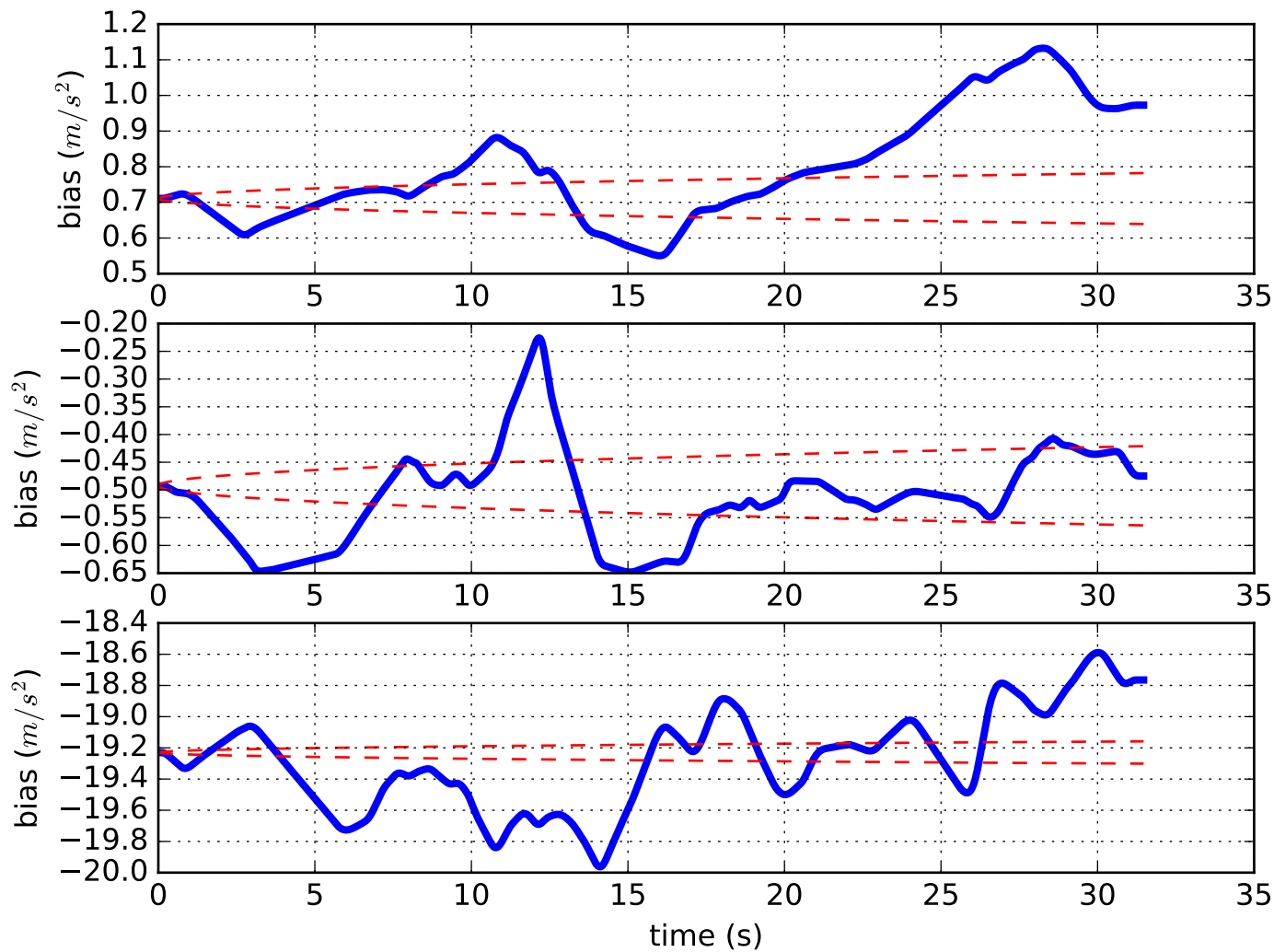
Comparison of predicted and measured specific force (imu0 frame)



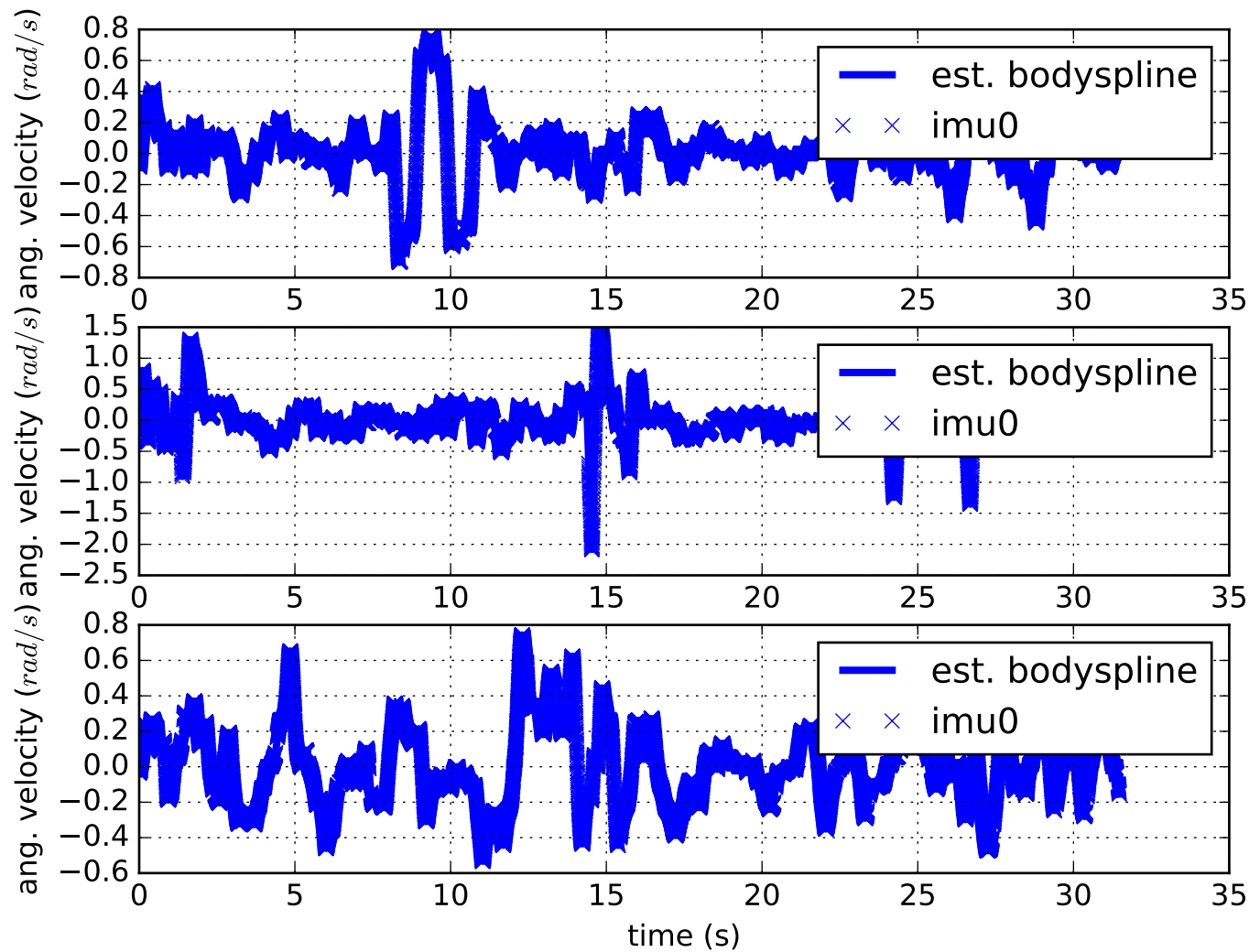
imu0: acceleration error



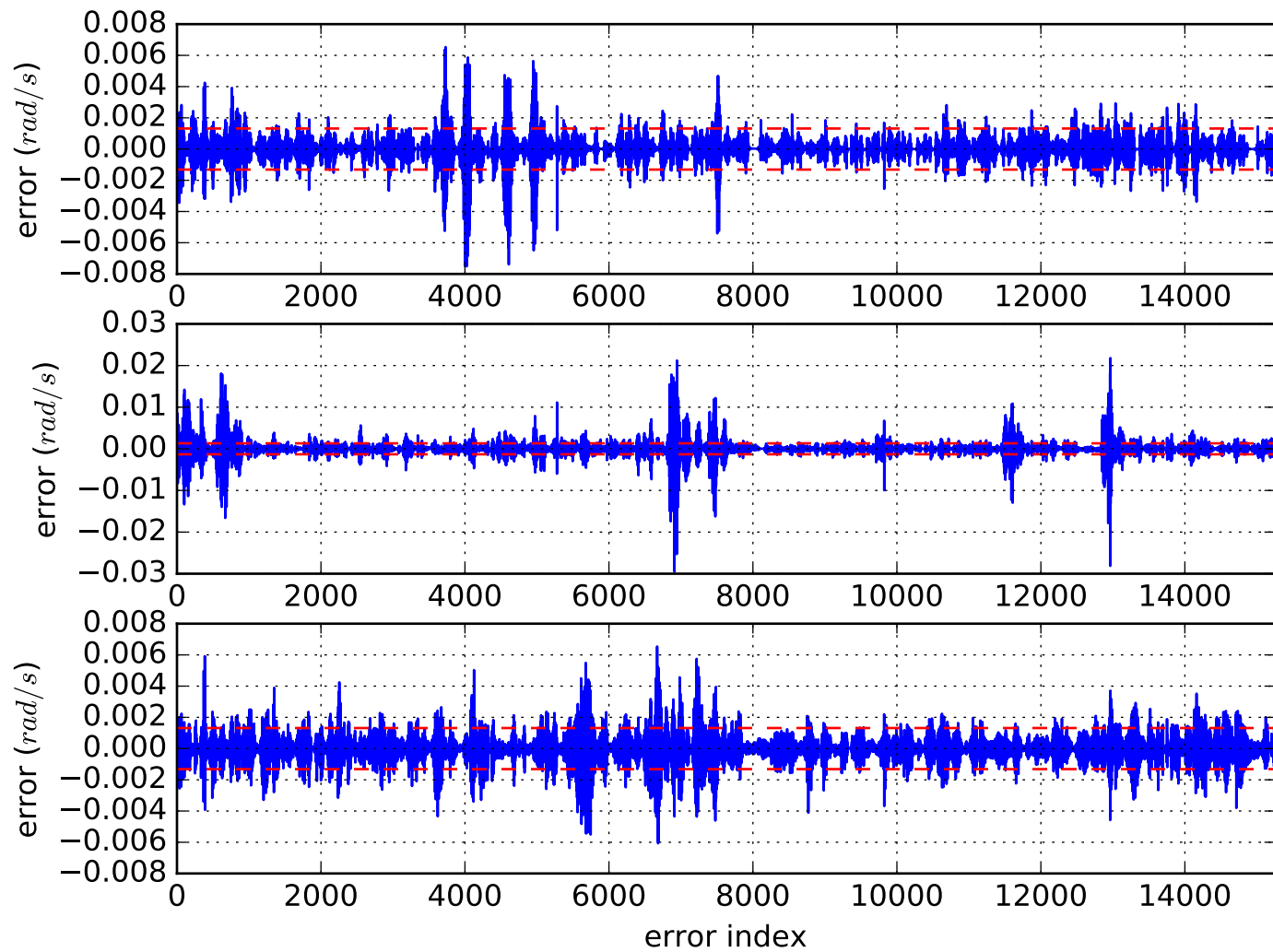
imu0: estimated accelerometer bias (imu frame)



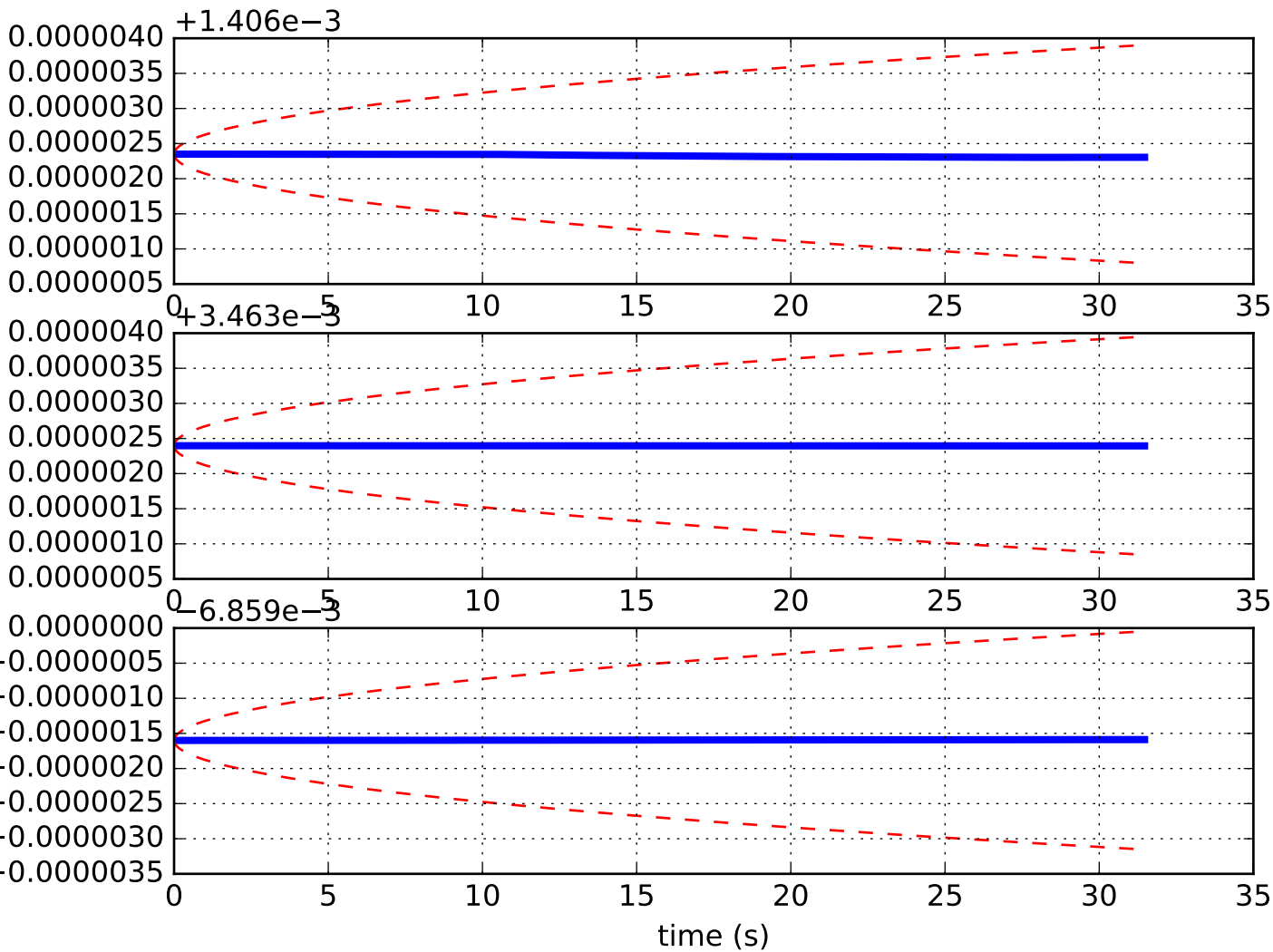
Comparison of predicted and measured angular velocities (body frame)



imu0: angular velocities error



imu0: estimated gyro bias (imu frame)



cam0: reprojection errors

